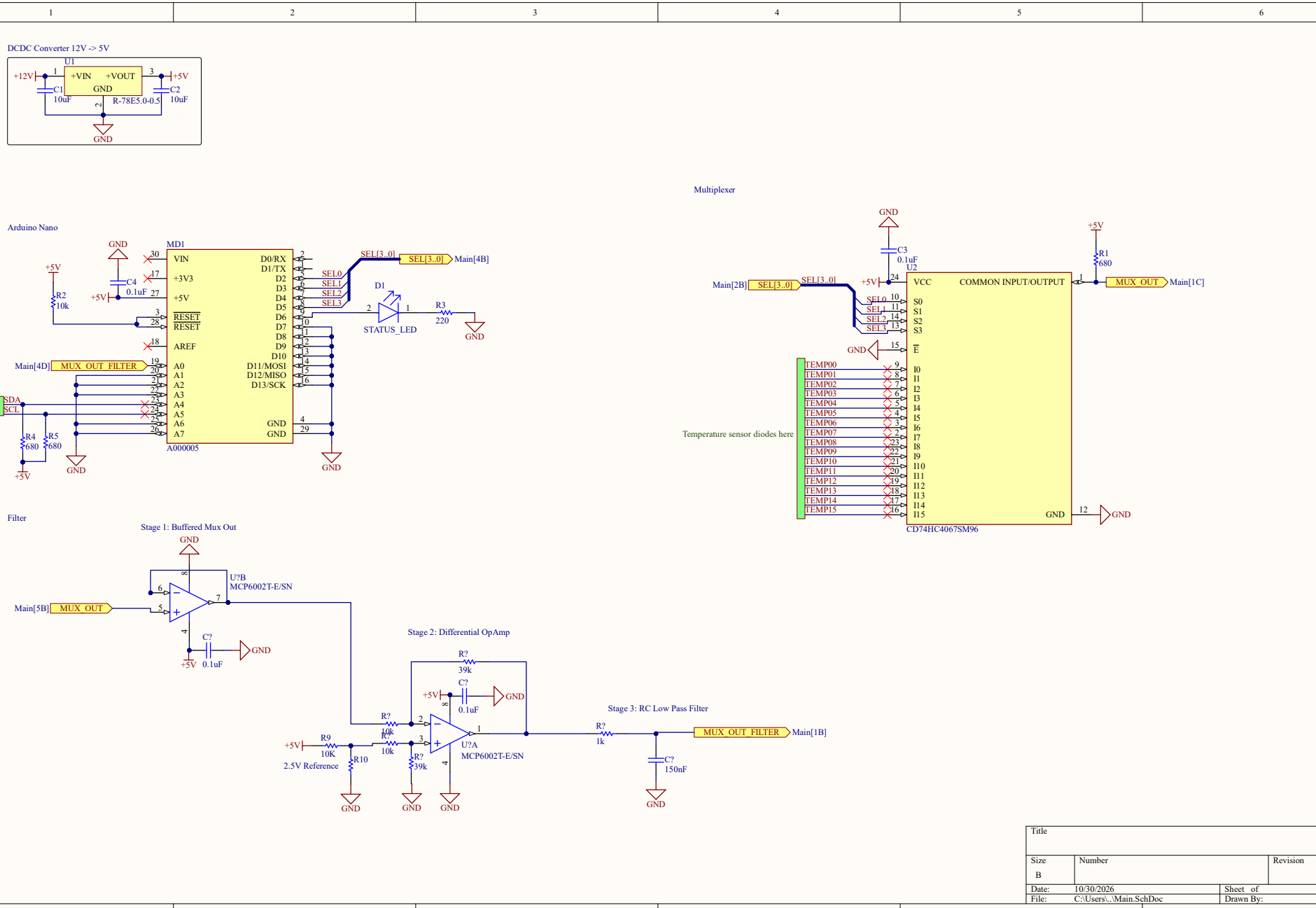

DCDC Converter 12V -> 5V
U1
+12V
+VIN
+VOUT
GND
+5V
C1
10uF
C2
10uF
R-78E5.0-0.5
GND
Arduino Nano
MD1
VIN
+3V3
+5V
RESET
RESET
AREF
A0
A1
A2
A3
A4
A5
A6
A7
D0/RX
D1/TX
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11/MOSI
D12/MISO
D13/SCK
GND
GND
A000005
GND
+5V
R2
10k
C4
0.1uF
SEL[3..0]
SEL[3..0]
Main[4B]
SEL0
SEL1
SEL2
SEL3
D1
STATUS_LED
R3
220
GND
Main[4D]
MUX_OUT_FILTER
I2C Output
SDA
SCL
R4
680
R5
680
+5V
GND
Multiplexer
GND
C3
0.1uF
+5V
R1
680
U2
Main[2B]
SEL[3..0]
SEL[3..0]
+5V
VCC
COMMON INPUT/OUTPUT
MUX_OUT
Main[1C]
SEL0
SEL1
SEL2
SEL3
S0
S1
S2
S3
GND
E
Temperature sensor diodes here
TEMP00
TEMP01
TEMP02
TEMP03
TEMP04
TEMP05
TEMP06
TEMP07
TEMP08
TEMP09
TEMP10
TEMP11
TEMP12
TEMP13
TEMP14
TEMP15
I0
I1
I2
I3
I4
I5
I6
I7
I8
I9
I10
I11
I12
I13
I14
I15
GND
GND
CD74HC4067SM96
Filter
Stage 1: Buffered Mux Out
GND
U?B
MCP6002T-E/SN
Main[5B]
MUX_OUT
C?
0.1uF
+5V
GND
Stage 2: Differential OpAmp
R?
39k
C?
0.1uF
+5V
GND
R?
10k
R?
10k
U?A
MCP6002T-E/SN
+5V
R9
10K
2.5V Reference
R10
R?
39k
GND
GND
GND
Stage 3: RC Low Pass Filter
R?
1k
MUX_OUT_FILTER
Main[1B]
C?
150nF
GND
Title
Size
B
Number
Revision
Date:
10/30/2026
Sheet of
File:
C:\Users\..\Main.SchDoc
Drawn By: